

Soil morphology related to hydrology and degradation in tropical coastal podzols (SE Brazil)

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ABSTRACT

Morphology of podzol profiles can be used to improve our understanding of soil formation and degradation processes. The morphology of a podzol chronosequence was studied for a 1800 m long cliff perpendicular to the coastline at the southern coast of the Island Ilha Comprida (São Paulo, Brazil). The large variation in hydrological conditions with time (surfaces with ages from 6000 to 300 yrs BP) and space (topography) allowed studying the relation between podzol morphology and drainage conditions, both during soil formation and degradation. On the oldest two surfaces, the podzols formed under poorly drained circumstances. Uplift during their formation resulted in (1) very deep Bh horizons and (2) degradation of the upper B horizon as a result of improved drainage. On the younger surface most podzols were poorly drained, but well-drained podzols and transitions occurred in beach ridges. In the cliffs on the southern coast of the island, improved drainage of the poorly drained podzols resulted in degradation of the B horizon. This degradation exhibits various forms, including (1) complete removal of the B horizon down to an underlying clay layer, (2) degradation of the B horizon along root channels with preferential vertical water transport, (3) degradation of the upper B horizon resulting in a diffuse EB transition, and (4) areas of organic matter (OM) depletion related to microbial activity, sometimes combined with burrowing. Type (2) degradation may result in a convoluted EB transition, the formation of islands of B horizon in the deepening E horizon, and formation of E horizon tongues into the Bh. Recent (post degradation) OM bands were found in E and B horizons. A thorough study of podzol morphology can be used to reconstruct several stages of their formation and degradation in relation to drainage conditions.

1. Introduction

Podzols have important environmental functions such as being a source of drinking water, a filter for pollutants and a sink for atmospheric carbon (Sauer et al., 2007). Therefore, it is important to improve our understanding of the processes involved in the formation and degradation of podzols. Several theories have been proposed to explain the formation of podzols (Anderson et al., 1982; Buurman and Jongmans, 2005; DeConinck, 1980; Lundstrom et al., 2000a). In all theories, the complexation and transport of dissolved organic matter (DOM) with Fe and Al plays an essential role in the formation of distinctive horizons. The morphology of podzols primarily reflects the processes of transport and accumulation of organic matter (OM) and Al and Fe compounds, i.e., podzols are typically composed of a shallow dark surface A horizon, followed by a grayish white eluvial E horizon from which OM, Al and Fe are removed, and a black to reddish brown

illuvial B horizon enriched in OM and Al and sometimes also Fe (Lundstrom et al., 2000b; Petersen, 1976).

Original drainage conditions can be deduced from the presence or absence of Fe. Poorly drained podzols have lost all Fe due to reduction and lateral removal, while in well-drained podzols Fe is still present and only removed from the E horizon. Intermediate drainage conditions are recognised by Fe mottles (Buurman, 1984; DeConinck et al., 1974). In poorly drained podzols the B horizon shows monomorphic OM coatings on quartz grains due to a source from DOM. In well-drained podzols the B horizon OM is both monomorphic (DOM coatings) and polymorphic (decayed root material) (Buurman and Jongmans, 2005). The presence of DOM coatings is evidenced by micromorphology, but may also be indicated in the field by a homogeneous colour and cementation of the B horizon. Drainage conditions cause distinctive macromorphological characteristics such as the presence or absence of Fe compounds features, the thickness of the B horizon, and the shape of the EB transition.

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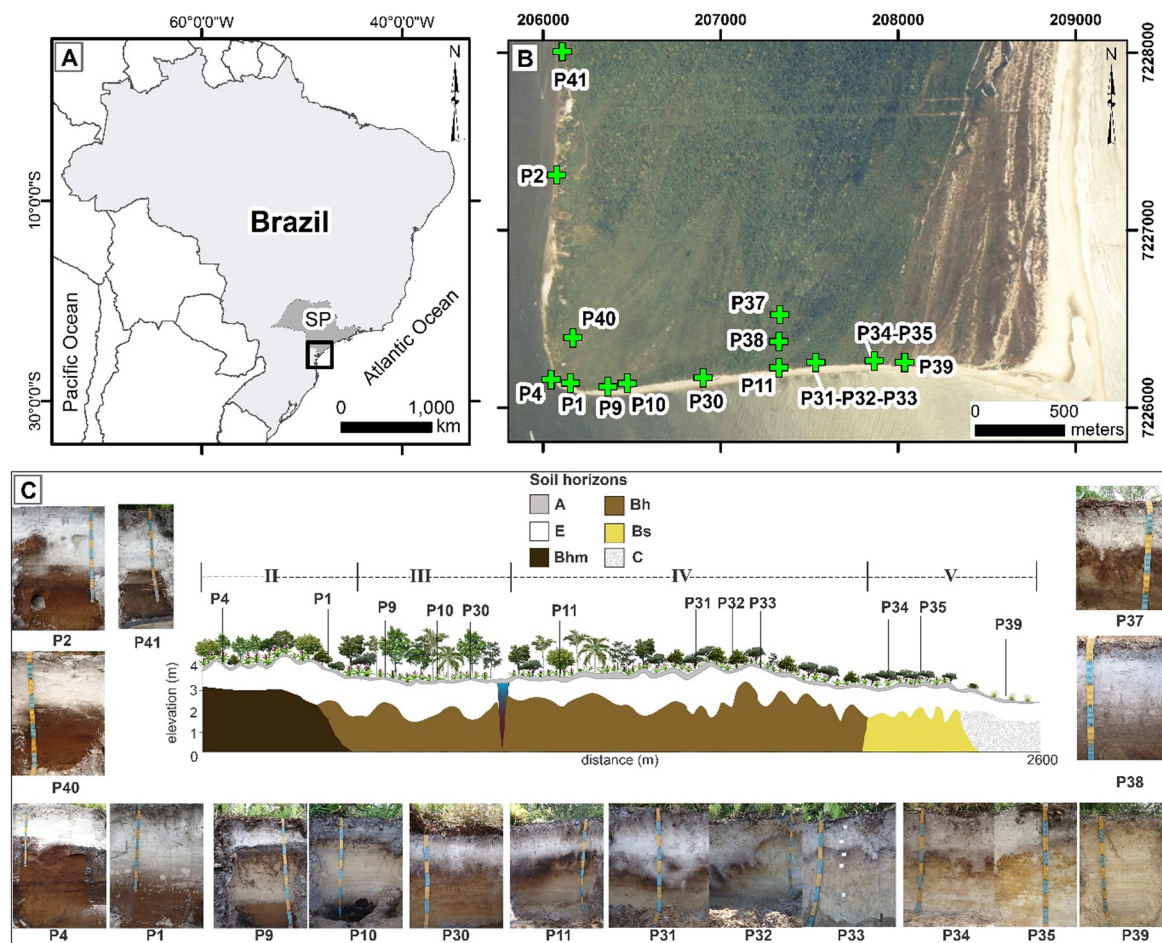


Fig. 1. Location of the investigated soil profiles on the sedimentary surfaces of Ilha Comprida. (from [Martinez et al., 2018](#)).

Poorly drained podzols, i.e. those (periodically) affected by groundwater, have lost Fe compounds because these have been reduced and removed by groundwater flow. Well-drained podzols, unless developed on pure quartz sands, generally have an accumulation of Fe compounds in the lower B horizon, while profiles in which the groundwater only occasionally reaches into the lower B horizon, may show redoximorphic features as mottles or concretions of Fe. Poorly drained podzols usually have a flat EB boundary caused by the highest groundwater level, while well-drained podzols have a thin undulating Bh horizon because shape and depth depend on vertical movement of percolating water. Other terms that are frequently used include groundwater podzols, hydromorphic podzols, (poorly drained), and xeromorphic podzols (well-drained), classified as carbic podzols and haplic podzols, respectively ([WRB, 1998](#)). For the nomenclature of different classification systems we refer to [Sauer et al. \(2007\)](#). Thus, podzols exhibit a morphology that is in equilibrium with drainage conditions during their formation. The effect of lateral changes in drainage on podzol morphology have been studied in detail ([Buurman et al., 2005, 2013; Kaczorek et al., 2004; Schwartz, 1988](#)), but the effect of changing hydrology over time has hardly been addressed. The morphology of poorly drained podzols may change as a result of improved drainage ([Buurman et al., 2005; Reuter, 1999](#)), and present morphology can therefore be used as the basis for studying (past) hydrological and chemical changes in podzols.

Podzolization processes in the tropics are similar to those in colder climates, but there may be essential differences in time of development and thereby climate stability. Due to the occurrence of ice ages, podzols in boreal and temperate areas are of Holocene age and have undergone only mild variations in climate. Podzols in tropical lowlands, on the

other hand, may be much older and have consequently undergone larger variations in environmental conditions (climate, vegetation). Detailed studies exist of individual podzol profiles from Brazilian coastal areas ([Coelho et al., 2010a, 2010b, 2010c, 2010d; Gomes et al., 1998a, 1998b, 2007a, 2007b; Rossi and Queiroz-Neto, 2001, 2002](#)), but the effects of time and variation in environmental conditions on podzols remain largely unknown. Two recent papers studied several podzol profiles in relation to the landscape position, which allowed the identification of environmental changes during their formation. At Ilha Barbosa (SE Brazilian coast), [Boski et al. \(2015\)](#) studied progradation rates of coastal barriers and time-dependent OM degradation processes using ^{14}C datings of podzol B horizons and their molecular composition, respectively. At Ilha Comprida, another Holocene barrier island on the SE Brazilian coast, [Martinez et al. \(2018\)](#) studied two cross sections perpendicular to the coastline and distinguished five Geomorphic Units, with estimated ages of 5250, 3660, 1435, 1180, and 325 years BP. In the youngest surfaces, morphological differences of the predominantly poorly drained podzols were related to actual hydrology and topography of ridges and swales, while in the older surfaces uplift in combination with converging groundwater flow caused hyper developed B horizons. Erosion of the southern coast, where one of the transects was located, caused formation of a cliff. Due to lowering of the erosion base near the cliff, groundwater levels subsided, and caused improved drainage of the profiles in the cliff.

These coastal podzols thus show morphological characteristics related to the original drainage during its formation, as well as degradation characteristics due to improved drainage. The 1800 m long cliff with its exposed profiles offers a unique opportunity to study the

Table 1

Characteristics and decay features found in the studied podzol profiles. The location of the profiles is given in Fig. 1.

Unit ^a	OSL age (y BP) ^a	Relief	Profile location	Drainage class ^b		Fe	(partially) Cemented B horizon	Thickness of the B horizon (cm)	B horizon degradation features				
				WD	PD				Diffuse EB boundary	Root pockets (tongues)	B islands in the E horizon	Mottle type ^c	Horizontal OM bands
I	5250 ± 860	Ridges and swales	Cliff		P2		x	> 165		x	x	b,e	x
			Inland		P40			> 75				b,c,e	x
			Cliff		P41		x	30				c,e	
II	3660 ± 360	Ridges and swales	Cliff		P1		x	> 180	x			a,b,c,e	
			Cliff		P4		x	> 220				a,b,d,e	x
			Cliff										
III	1435 ± 155	Flat	Cliff		P9			37					
			Cliff		P10		x	40					
			Cliff		P30			65	x	x		a,b,e	
IV	1180 ± 92	Ridges and swales	Cliff		P11			> 100	x	x		b,e	x
			Cliff		P31			25		x	x	b,d,e	
			Cliff	P32		x,m		35		x	x	b,e	
			Cliff	P33		x,m		35		x			
			Inland		P37			40		x		c,e	
			Inland		P38			60	x			a	
V	325 ± 31	Flat	Cliff	P34		x		30	x	x	x	b,e	
			Cliff	P35		x,m		20	x	x	x		
			Cliff	P39				–					

^a OSL = Optical Stimulated Luminescence, from Martinez et al. (2018).

^b WD = Well drained; PD = Poorly drained; x = Fe is present according to analysis (SI_Table 1); m = Fe mottles that are visible in the field.

^c For a description of the different mottle types see Table 2.

Table 2

Types of OM depletion mottles found in Ilha Comprida Podzols (see Fig. 4 for representative photographs of mottles).

Type	Morphological characteristics	Occurrence
a	Concentric. With bands of more and less depleted zones. Usually with rounded top and flat bottom (stratified), but fully round forms also occurred. Mottles did not merge. Maximum observed size about 20 cm.	In upper part of Bh horizons of poorly drained podzols.
b	Circular/tubular. 1–4 cm across. Large contrast with surrounding. They sometimes merged. Sometimes oriented along sedimentary contacts.	Most common in EB and BE horizons of poorly drained podzols.
c	Dotted. Small, circular. Occurred in large concentrations sometimes.	Most common in the lower Bh horizon of poorly drained podzols of Unit II.
d	Ghost-like OM depletions. Large (up to 20 cm), irregularly shaped. Top has strongest bleaching. Not influenced by sedimentary structures.	In all Bh horizons of poorly drained podzols.
e	Irregular. Similar to type b, but larger.	In all Bh horizons of poorly drained podzols.

effects of time and drainage on podzol morphology. In the present paper, we focus on field morphology of these podzols in relation to their (changes in) drainage position, using chemical analysis and micro-morphology. This systematical and detailed study of decay features in relation to hydrology will further serve as a basis for the interpretation of organic geochemistry and ecological diversity.

2. Materials and methods

2.1. Study area

The east coast of Ilha Comprida (São Paulo State, Brazil) is

approximately 74 km in length, and its width varies from 625 m to 5.4 km (Gerardi and Mendes, 2001). Ilha Comprida consists predominantly of sandy sediments in the form of ridges, which are geomorphic expression of a prograding barrier. Soil profiles were studied in the cliffs along the 1800 m long south coast and the west coast of the southern end of Ilha Comprida (Fig. 1). The direction of the south coast is perpendicular to the Atlantic coast, and thus exposes a sequence of ridges. For more detail on the geomorphology of the Island we refer to Martinez et al. (2018). The climate is humid tropical (Af according to the Köppen classification), with a mean annual temperature of 23 °C, and mean annual precipitation of 2130 mm (Alvares et al., 2013). Rainfall is well distributed throughout the year.

2.2. Sampling and morphology

The morphology of 42 profiles was studied in the field, including profiles from the cliffs and at locations 50–100 m inland. Seventeen of these profiles were selected for detailed description and sampled for chemical analysis. This selection of profiles was representative for the different geomorphological units and field morphology of the profiles (Fig. 1). The inland profiles were intended to eliminate the effects of wind, marine spray and tide, to which the podzols in the cliff are exposed. These inland profiles should also show less influence of the lowered erosion base on local hydrology. All profiles were described according to Santos et al. (2013). The seventeen profiles were sampled according to pedogenetic horizons, resulting in a total of 144 samples for chemical analysis.

2.3. Chemical analysis

Chemical analyses included pH (H₂O); Ca²⁺, Mg²⁺ and Al³⁺ exchangeable by KCl 0.1 M (McLean et al., 1958); H + Al by calcium acetate; P, K⁺ and Na⁺ by Mehlich (1953). Extraction with 0.1 M sodium pyrophosphate was done at pH 10 according to USDA (1996) and reflects the Fe and Al that is bound to OM (Fe_p and Al_p). Organic carbon (C_{org}) was determined by wet oxidation with potassium dichromate (K₂Cr₂O₇) 0.4 M (Walkley Black).

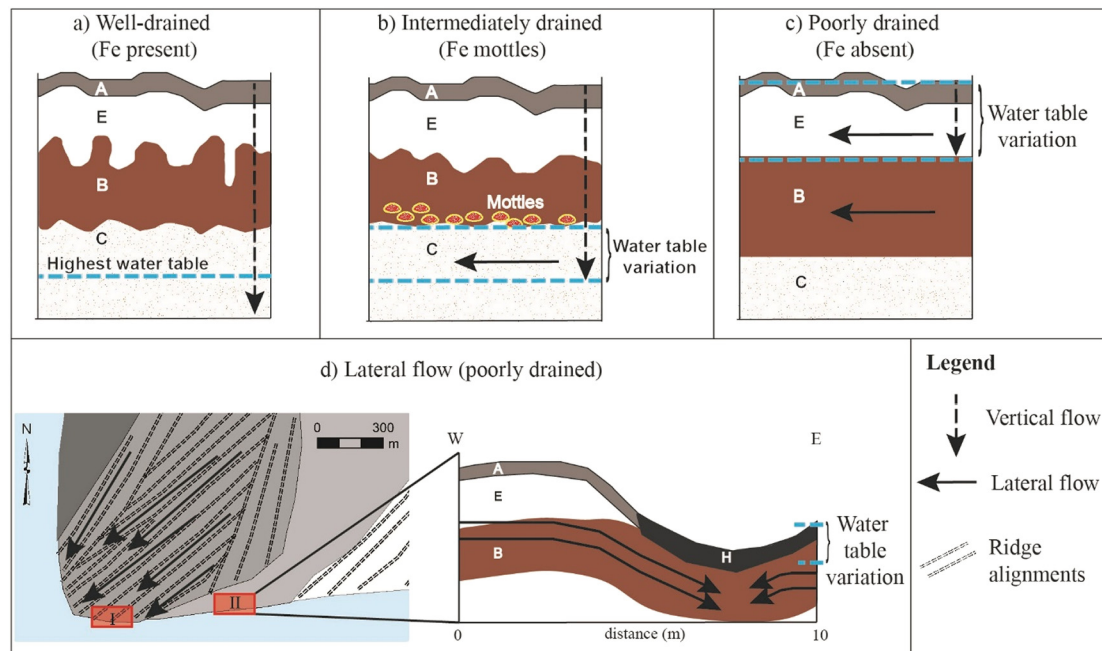


Fig. 2. Schematic overview of (a–c) groundwater level and water flow for the main drainage situations, and (d) drainage area for poorly drained podzols, including regional (I) and local (II) lateral flow.

2.4. Micromorphology

A total number of 23 undisturbed samples were collected in the field for micromorphology, using cardboard boxes with dimensions of $12 \times 7 \times 4$ cm. Selection was representative for the study area and focussed on horizon transitions and decay features. Thin sections were prepared according to Castro et al. (2003). They were analysed using a trinocular petrographic microscope coupled to a colour CCD camera from the Optical Microscopy Laboratory of the Department of Soil Science ESALQ/USP (Piracicaba, Brazil). The micro-features were described for 4 representative thin sections according to the criteria and terminology of Bullock et al. (1985), and humus forms according to DeConinck et al. (1974).

3. Results and discussion

3.1. Geomorphology and soil classification

Podzol profiles were classified as well-drained or poorly drained according to the presence of Fe (SI Table 1). Drainage condition and position in the field (inland/cliff) of the profiles, and age and relief of the Geomorphical Units on which they were found (Martinez et al., 2018) are summarized in Table 1. In short, the oldest surfaces (Units I and II) exhibited poorly drained podzols with strongly developed B horizons. The third surface (Unit III), which is rather flat, has peats and poorly drained podzols with less pronounced B horizons. The fourth surface (Unit IV) contains both poorly drained and well-drained podzols in short-distance sequences related to swales and ridges, while the youngest surface (Unit V) has a narrow zone of well-drained podzols at its landward side and Arenosols further seaward (Martinez et al., 2018). Erosion of the southern and south-western coast of the island and gradual uplift of the SW part caused a steep decline in groundwater level close to the coast, which was not observed about 1 km inland (Martinez et al., 2018). This decline in groundwater level caused improved drainage of the profiles in the cliff. The podzol profiles in the cliff may thus show morphological characteristics related to both the original drainage during its formation and subsequent improved drainage.

3.2. Field morphology

The morphological and chemical characteristics of poorly drained and well-drained podzol B horizons are summarized in Table 2A (see Section 1), and their hypothesized groundwater levels and water flow are given in Fig. 2a–c. The thickness of the B horizon of poorly drained podzols is largely influenced by the amount of groundwater passing through the profile vertically and laterally, which in turn is determined by the landscape morphology. If water flow is local, the DOM in the B horizon will be limited to local litter and will not show very strong OM accumulation (Fig. 2d situation II), while external lateral flow of water may contain DOM from litter of a much larger area and thus be darker and have more OM accumulation (Fig. 2d situation I; Buurman, 1984). Morphological characteristics that relate to improved drainage of originally poorly drained podzols are summarized in Table 2B. They will be used in the description of the profiles (Sections 3.2.1–3.2.5) and include diffuse EB transitions, root pockets, recent OM bands, and mottles of microbial OM depletion. While transitions between E and B horizons in poorly drained podzols are usually both clear and horizontal (flat), upon improved drainage, the top of the B horizon may lose OM and the EB boundary may become gradual so that EB and BE horizons can be distinguished. Improved drainage may cause dominance of vertical over lateral water movement. Such water movement usually follows zones of larger porosity, such as around root channels, causing the formation of tongues or root pockets at the EB transition.

Improved drainage and downward water movement may cause the formation of OM bands overprinting the (partially degraded) Bh horizon. Such bands may be horizontal, and influenced by sedimentary stratification (Wiechmann, 1975), or follow tonguing root channels. They must therefore have formed more recently than the original B horizon. Well-defined mottles with sharp boundaries inside which OM has been depleted have been ascribed to microbial decay in well-drained Dutch podzols (Buurman et al., 2007). Such local depletions occurred in diverse shapes that are listed in Table 2 and illustrated in Fig. 3. The morphology of the profiles will be discussed by Geomorphical Units (Units I–V; Fig. 1).

3.2.1. Unit I (5250 y BP)

Profile P2 (Fig. 4) showed a homogeneous brown B horizon with

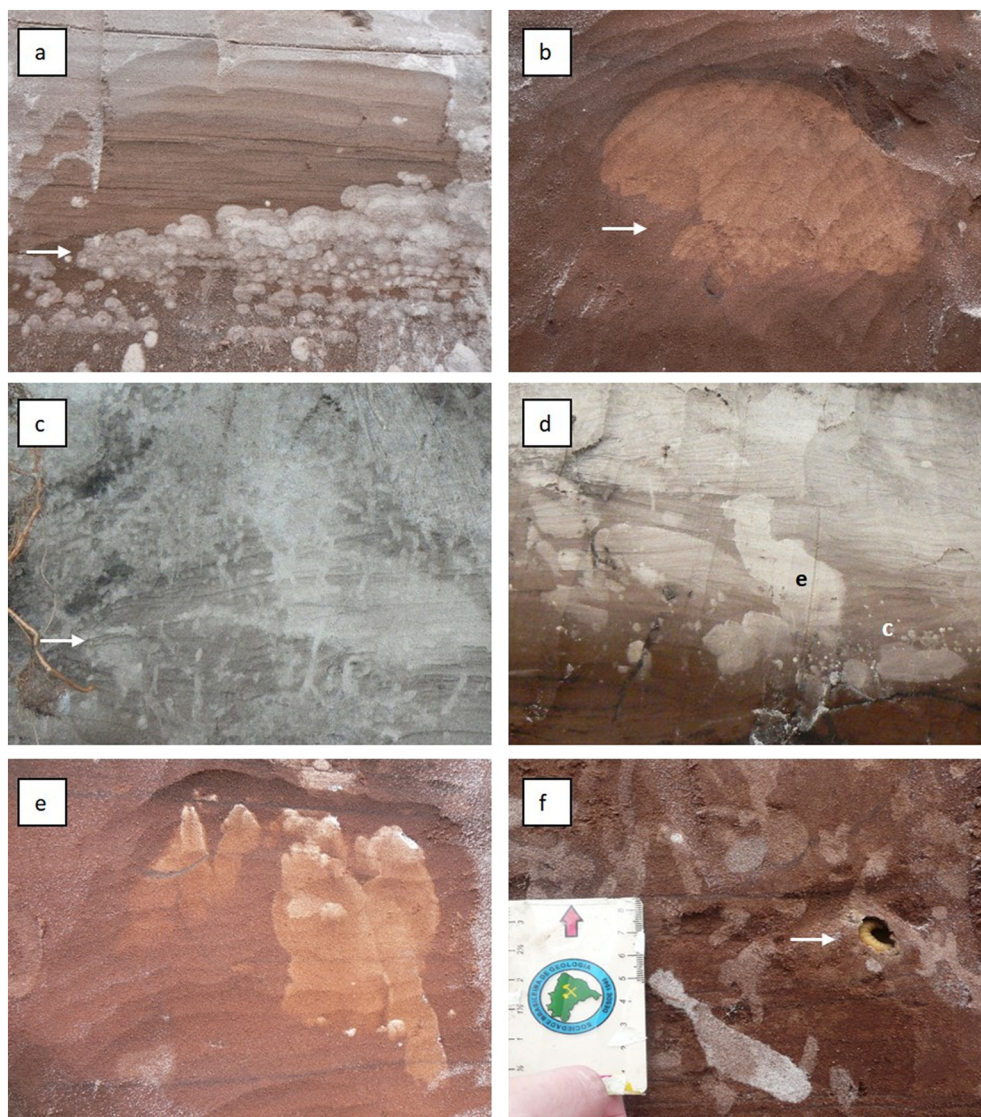


Fig. 3. Types of OM depletion mottles (see Table 2 for a description of the mottle types). (a) Type a concentric mottles in the upper B horizon of P1, between 180 and 200 cm depth; (b) Large concentric type a mottle from profile P4 at 180 cm depth; (c) Type b circular/tubular mottles in the EB horizon of profile P2; (d) Irregular type e mottles at the EB transition in profile P2. Note that the stratification appears unaffected by the depletions; (e) Ghost-like type d mottles at 160 cm depth in profile P4; (f) An immature individual of Coleoptera, probably *Scarabaeidae*, found in a cave in the cliff. Note the lighter colours of the burrows.

very clear degradation features. An originally horizontal upper boundary can be recognised (Fig. 4c), but locally only remnants of B horizon remain, surrounded by E horizon (Fig. 4a). In combination with the absence of Fe (SI Table 1), this indicates that this profile was formed under poorly drained conditions, and that subsequent improved drainage has resulted in degradation of the B horizon. Several features support improved drainage. (1) In some areas, the E horizon has penetrated the B horizon in the form of root pockets (Fig. 4b,c), while in other areas, almost the whole B horizon has disappeared down to the underlying clay layer (Fig. 4d). (2) Horizontal OM bands were present that continued through both the original B horizon and root pockets (Fig. 4a,b). Fig. 4b shows a root remnant in the centre of the channel, surrounded by a bleached horizon in which OM bands that follow the shape of the root pocket are evident. (3) All types of microbial depletion mottles were present both in the E and B horizon.

Profile P40 (Fig. 5a) is from an inland profile pit, 1.5 m deep. This shallow profile did not reach the clay layer that was found in P2 (Fig. 4d). Similar to profile P2, it has a homogeneous thick B horizon (at least 70 cm) devoid of Fe, indicating a poorly drained profile. The diffuse upper boundary of the B horizon in combination with microbial mottles in the EB and upper B horizon (types b,c,e) indicates degradation of the B, which point towards improved drainage, but these features are less developed than in P2.

Profile P41 contained two superposed podzols separated by a

sedimentary layer (Fig. 5b). The lower buried B horizon was cemented and more than two meters thick; the absence of E and A horizons above the B of the buried profile is explained by marine erosion. In the overlying sediment, a new podzol developed, but with a B horizon of only 30 cm thick. Both podzols showed the characteristics of poorly drained profiles (devoid of Fe, flat upper boundary of the B). The upper podzol had no decay features, and thus no evidence for improved drainage.

3.2.2. Unit II (3660 y BP)

Both profiles P1 and P4 both showed typical characteristics of poorly drained groundwater podzols, with a giant, homogeneous B horizon that is devoid of Fe and partially cemented. The EB boundary showed some root pockets, but is predominantly horizontal. Both profiles are located in the highest cliff of the south-west corner, indicating a major impact of improved drainage. In profile improved drainage is indicated by a diffuse EB transition (Fig. 5c), and abundant microbial mottles (a, b, and e-type) till deep in the Bh horizon. In profile P4 (Fig. 5d), the homogeneous Bh horizon reaches a depth of > 4 m. Much of the B horizon is cemented. The original upper boundary was flat and abrupt, but is now affected by root penetration that caused a wavy to tonguing boundary at some places. The top of the B horizon has a, b, and d-type depletion mottles.

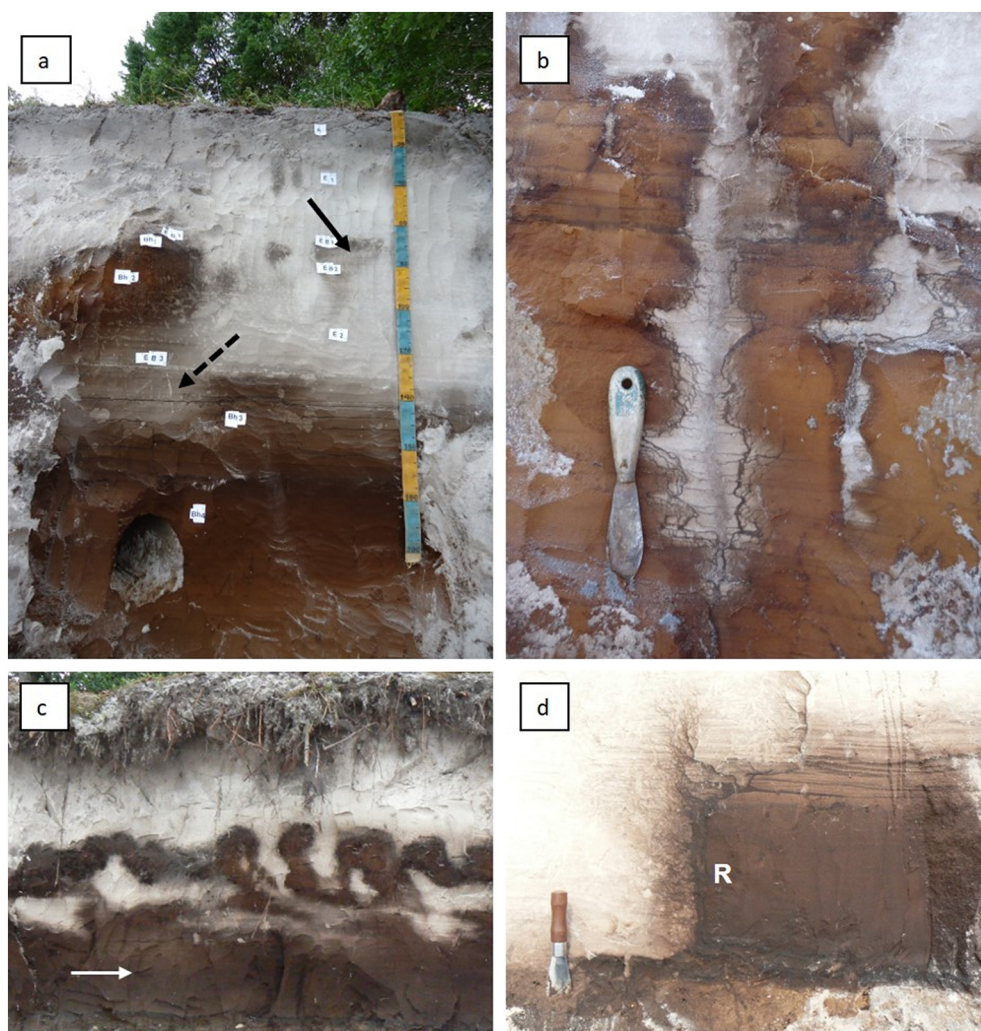


Fig. 4. Profile P2. (a) Convoluted top of B horizon with Bh remnants surrounded by E (arrow). Dashed arrow points to recent OM band. (b) Disappearance of the B horizon along a root channel. Note OM bands that follow the shape of the channel and other ones that are horizontal. The decaying root is still visible. (c) Strongly convoluted top of the profile due to degradation of the B horizon along roots. The fact that the EB transition is not affected indicates that these shapes are not due to tree throw. (d) Deepening of the E horizon on the left has removed part of the original poorly drained B (homogeneous brown colours) on the right. In the center more recent OM accumulation, presumably starting from a vertical root channel (R). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2.3. Unit III (1435 y BP)

The surface of unit III is almost flat, and consequently all profiles have the same drainage pattern. In shallow depressions, podzols with a peat cover were found (Martinez et al., 2018), but only profiles without peat covers were sampled.

Profile P9 (Fig. 6a) was virtually devoid of Fe compounds and therefore classified as a poorly-drained podzol. It has a well-developed A horizon and about 20 cm of E horizon overlying a thin dark Bh horizon. Under this thin Bh horizon is a slightly bleached horizon, under which a second (weak) Bh horizon was present that has many root remnants extending into the C horizon. The abundant roots and the division of the Bh suggest that upper part of the present Bh horizon might be a buried A horizon that later became part of the B horizon of the overlying profile. Below the stratified C horizon, at 170 cm depth, is the remnant of the a poorly drained B horizon from a podzol that lost A and E horizons through erosion. Apart from erosion, this profile lacks evident degradation features.

Profile P10 (Fig. 6b) is a weakly developed poorly drained podzol. The A, E, and B horizons together hardly exceed 40 cm, of which the Bh is only a few cm thick. Below the Bh are some decayed roots, and the Bh itself is strongly rooted. The profile developed in strongly stratified beach sands that contain, below 160 cm, rounded blocks of podzol B material that originate from erosion of a previous poorly drained podzol (Martinez et al., 2018). The profile does not have clear features that are related to improved drainage.

Profile P30 (Fig. 6c) is a poorly drained podzol. The B horizon is more developed than that of profile P9 and P10. There is no macro-

morphological evidence of DOM accumulation through groundwater flow, and the Bh horizon gradually fades downward, its top being darker than the lower part. Degradation features include tongues, a gradual to diffuse EB transition, and microbial mottles (types a,b,e).

3.2.4. Unit IV (1180 y BP)

Unit IV has well-defined beach ridges and swales and podzol development under different hydrological regimes. Profile P11 (Fig. 6d) is a weakly developed profile in a swale that is completely devoid of Fe compounds, and thus poorly drained, but it does not have macro-morphological evidence of DOM accumulation through groundwater. There are many thin and thick roots in the O horizon and a few in the A horizon. The vegetation undergrowth consists of Bromeliads. There are many irregular bands of (downward) DOM accumulation in the Bh₁ and Bh₂ horizons and a few in the Bh₃ horizon. Many dark remnants of decomposing fine roots are found mainly in the Bh₁ and Bh₂ horizons. The latter has larger and distinct root channels filled with decomposed root material. The transition between the E and Bh horizons is irregular and in some areas in the form of tongues. The transitional EB horizon presents many type c mottles. In the Bh₃ horizon, at the base of the profile, there is a clear parallel sedimentary stratification, which was not so visible in other horizons.

Profiles P31–P33 form a sequence from a swale to a ridge (Fig. 7a). In the swale, poorly drained profile P31 is devoid of Fe compounds and (Fig. 7b) has E horizon tongues, that suggest a periodical vertical passage of water (due to its lower position), but without DOM accumulation through groundwater because the lower part of the B horizon is not

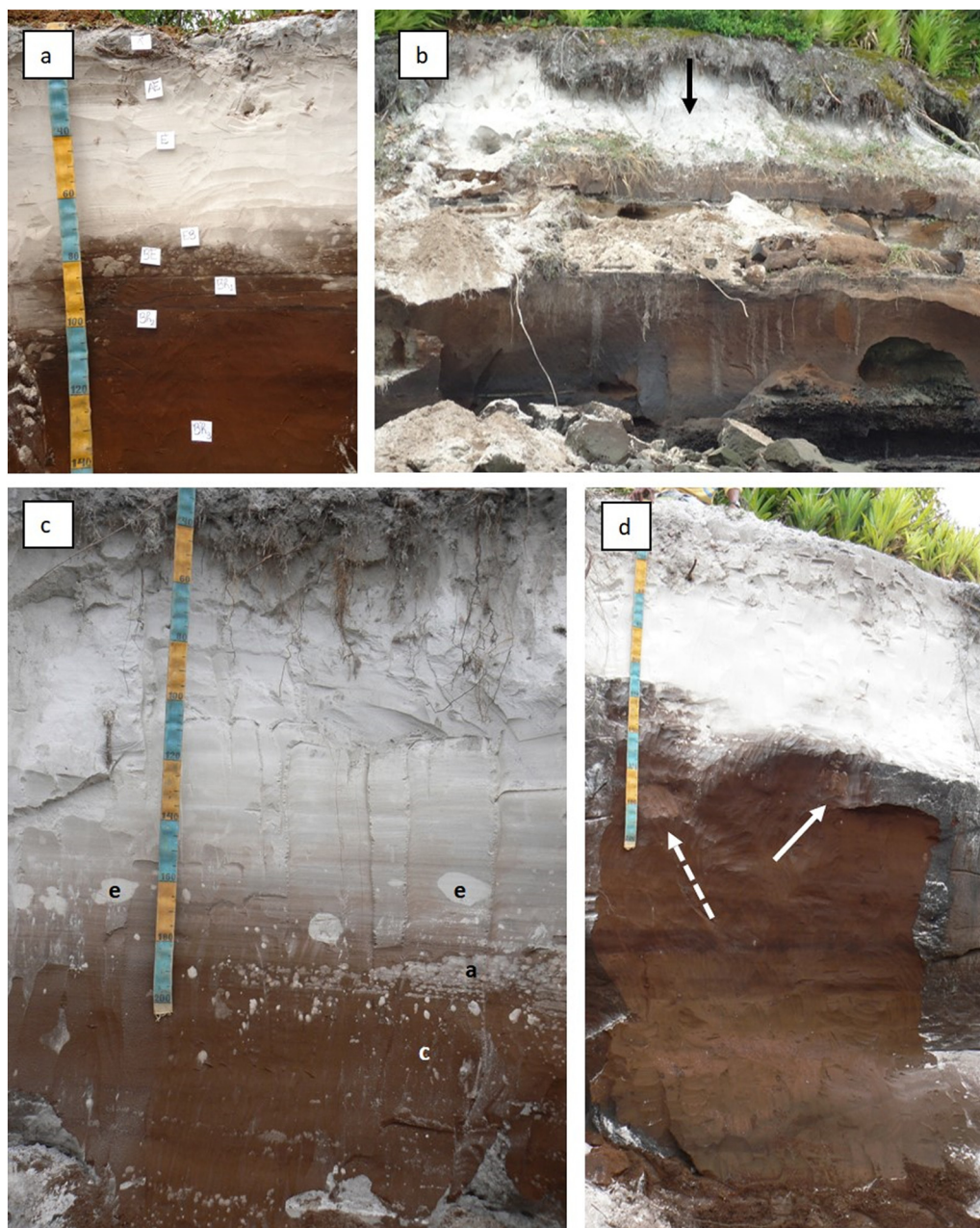


Fig. 5. (a) Profile P40. Note the depletion mottles at the EB transition. (b) Profile P41. The upper profile (arrow) was investigated. The bottom profile is similar to profiles P2 and P4. (c) Profile P1. Note the diffuse EB transition and large OM depletion mottles between 160 and 180 cm depth. The tape is 2 m long. Letters refer to mottle types (Table 1). (d) Profile P4. The arrow points to ghost-like type d mottles; dashed arrow points to concentric type a mottle. The tape is 2 m long.

homogeneously coloured by OM. The vertical passage of DOM-laden water has caused a well-developed Bh horizon. In the lower portion of the profile, the sediment is stratified and bands with dark minerals are visible.

Profile P32 (Fig. 7c) is located in the middle portion of the transect, in between the beach ridge and swale. Presence of Fe compounds indicates that the profile has good drainage. It has more abundant roots in the surface layers than profile P31. Some areas in the lower horizons have an accumulation of decomposing very fine roots. The profile has E and Bh horizons in the form of tongues, which presents a very different morphology compared to the lowest (P31) and highest (P33) part of the transect. The bleached parts of the tongues are inclined towards the ridge, which may indicate outward flow of water from the swale caused by lateral pressure during high water. The Bh₁ horizon near the swale (left hand part of Fig. 7c) is deeper. It is dark and rather homogeneous colour suggests a DOM source. Towards the right, the B horizon (coded Bh') presents a discontinuous wavy line. The Bh₁' horizon can be clearly differentiated from the bottom Bh₁ horizon because there are

concentrations of Fe hydroxides in the right hand portion of the profile, nearer to the ridge. These concentrations appear as small (2–6 mm) and hardened reddish spots and are thus somewhat influenced by ground-water. In most of the profile, these Fe concentrations were associated with root channels. These channels are more bleached and have a high concentration of fine root remains. Where the Fe remains, roots are rare or absent. The region below the Bh₁' horizon has a yellowish colour due to Fe compounds and was designated Bs. The lower portions of this section have a C horizon in which thin bands of dark sand can be distinguished.

Profile P33 (Fig. 7d) is located in the highest part of the landscape (ridge). The profile has good drainage, with abundant roots in the surface horizon and some concentrations of fine roots in the subsurface horizons. The profile is weakly developed, with a thin and wavy, sometimes tonguing Bh₁' horizon. The yellowish colouration of the subsurface horizons reflects that this profile contains larger amounts of Fe compounds (up to 1.4 g kg^{-1} in the Bh' horizon; SI Table 1) than profiles P31 and P32. There are only a few iron mottles in the left hand

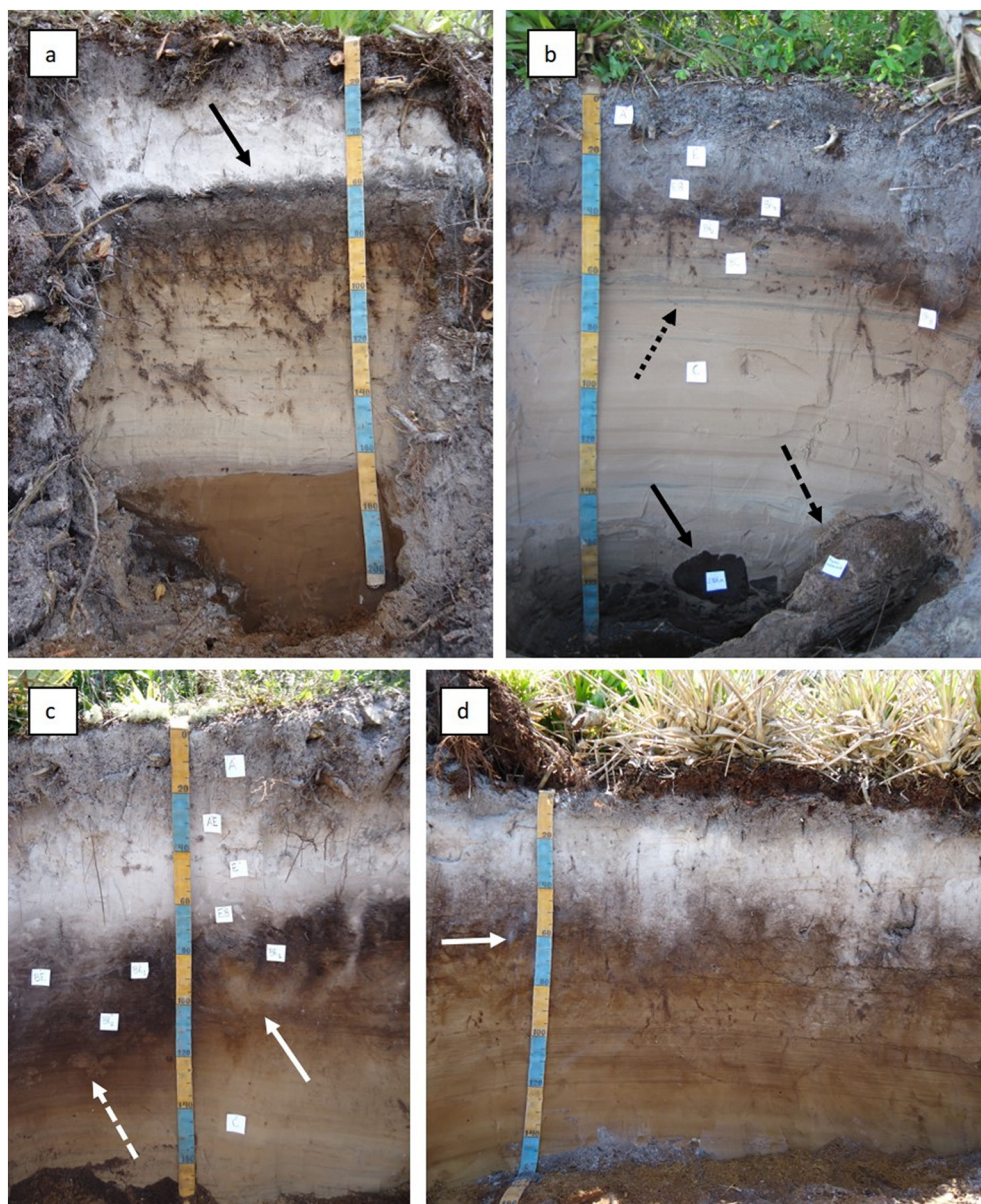


Fig. 6. (a) Profile P9. The top of the Bh horizon might be a buried A horizon. Note extensive rooting in and below the Bh. At 160 cm is the eroded Bh horizon of an earlier groundwater podzol. (b) Profile P10. Note bands of heavy minerals (dotted arrow), rolled Bh horizon remnant (arrow) and buried tree trunk (dashed arrow). (c) Profile P30. Note Fe mottles at right of tape (arrow) and concentric type a depletion mottles on the left (dashed arrow). (d) Profile P11. Weakly developed profile with some recent OM bands.

portion of the profile, indicating that the highest part of the profile was never influenced by groundwater. The C horizon has many remnants of root channels that (1) have lost Fe compounds, and (2) contain decayed root remnants. The different appearance of Fe is illustrated for profile P35 (Fig. 8d).

Decay features are clearest in profiles P31 and P32. In the Bh₂ horizon of profile P31, both type a and d mottles are encountered. In profile P32, part of the B horizon has been degraded and locally only 'islands' remain (Fig. 7e), surrounded by bleached sand. In many places, the top of the Bh horizon (EB, BE) has become diffuse and it contains many type b bleached areas. The well-drained profile P33 does not show clear degradation features.

The inland profiles P37 and P38 are practically devoid of Fe compounds and therefore poorly drained. Profile 37 (Fig. 8a) has a Bh horizon of 20–30 cm thickness with wavy upper and lower boundary. Remnants of roots are abundant below the B horizon. The E-horizon is well-developed and about 40 cm thick. There are some E horizon tongues into the B horizon. The EB transition locally has (fused) type e mottles up to 5 cm in diameter. Lack of homogeneous colour in the lower B indicates absence of DOM accumulation through lateral water flow. In profile P38 (Fig. 8b), however, the Bh horizon is

homogeneously brown, suggesting that it may have formed that way. The Bh horizon of P38 has many type c mottles, which appear to fuse to type e in the EB transition, suggesting that the E is affected by microbial decay. Both the E and B horizons have decaying roots.

3.2.5. Unit V (325 y BP)

On Unit V, podzol profiles are exclusively found at the transition to Unit IV. Morphologically, profile P34 (Fig. 8c) is similar to P31, but notwithstanding its younger age, it has a larger content of Fe compounds (up to 0.5 g kg^{-1} ; SI Table 1), suggesting better drainage. The Bh horizon is only a few cm thick and very wavy. In some places, the EB horizon is rather thick, suggesting a degradation of the B horizon. Both the Bh and the EB/BE horizons have type b OM mottles.

Profile P35 (Fig. 8d) shows the transition to a well-drained podzol. This profile has Fe-compounds (up to 1.3 g kg^{-1} ; SI Table 1) that appear affected by gleying in the C horizon. In part of the top of the B horizon, Fe concretions are found, which indicate that groundwater incidentally reaches this high. The B horizon has very little OM accumulation. Breakdown of the B horizon is illustrated by islands of B horizon surrounded by E horizon. Type b mottles occur at the EB transition, but they are much less clear than in the other profiles.

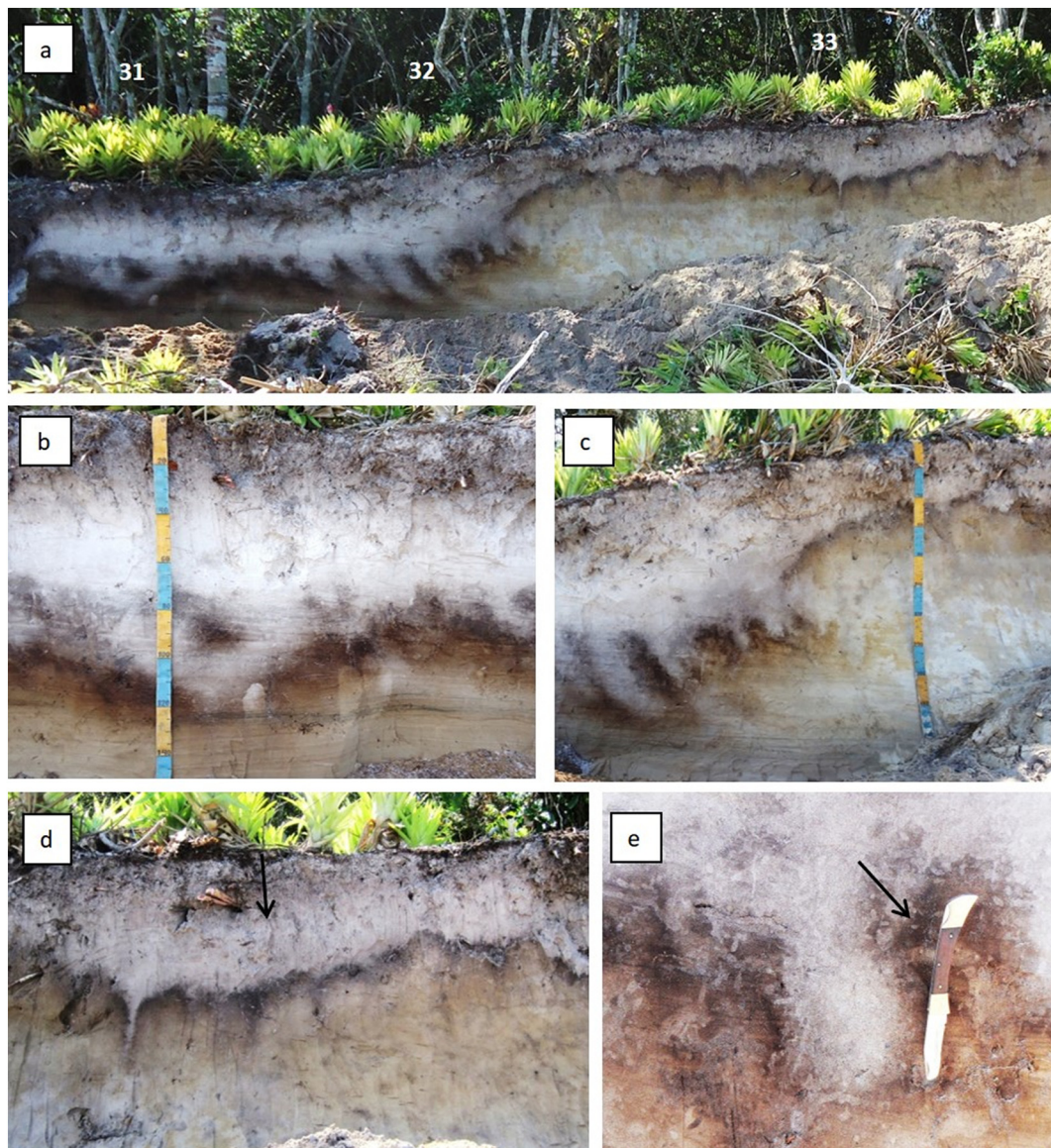


Fig. 7. (a) The hydrosequence of profiles P31–P33. (b) Profile P31. Note the ‘islands’ of Bh horizon in the E. (c) Profile P32 with inclined E horizon tongues. From the arrow towards the right, the B horizon contains Fe (Bh’). (d) Profile P33 with thin Bh horizon and diffuse Fe compounds below the Bh. Note roots. (e) Profile P32. Island of Bh horizon (arrow) and abundant burrow-like type b mottles at the EB transition.

Profile P39 (Fig. 8e), finally, is located on the seaward side of the podzol belt in unit V. It is a Psamment profile that has no evidence of podzolization.

3.3. Micromorphology

As expected, the Bh horizons of poorly drained podzols (represented by profiles P2, P4, P31 and P38) contained monomorphic organic coatings (Fig. 9a,b), demonstrating a DOM source. The fact that in many places these coatings completely filled the pores may explain the cementation of these horizons. In the EB horizon (as illustrated for profile P2 in Fig. 9c,d) OM coatings were thinner, which is interpreted as a result of (microbial) degradation (Buurman et al., 2005). In the poorly drained profile P31, monomorphic coatings dominated the B horizon (Fig. 10a,b), but OM pellets related to decaying roots were also encountered. Thus, the OM in these Bh horizons have a contribution from both DOM and decayed roots. Monomorphic coatings in this profile hardly ever completely filled pores, which explains why these Bh horizons are not cemented.

In the poorly drained and rooted inland profile (P38),

micromorphology showed excrements from mesofauna (pellets, polymorphic OM; Phillips and FitzPatrick, 1999) in and around decaying roots (Fig. 10c,d), and sometimes these pellets merged into dark coatings. The difference with the monomorphic coatings is that these dark coatings were not translucent, suggesting that they are features of degradation of pellets and not of DOM, in agreement with a large contribution from (decayed) roots to OM in podzol Bh horizons (Buurman et al., 2005; Buurman and Jongmans, 2005; Phillips and FitzPatrick, 1999).

3.4. Synthesis

Exclusively poorly drained podzols were found in Units I to III. Both poorly drained and well-drained podzols were found in Unit IV where, due to narrow swales and ridges, transitions between the two occur at a distance of a few meters only. In Unit V, only the landward fringe showed poorly developed, well-drained podzols while the seaward side did not show podzolization. Thick poorly-drained Bh horizons were restricted to Units I and II.

In Unit II, the thick Bh horizon was due to concentrated lateral

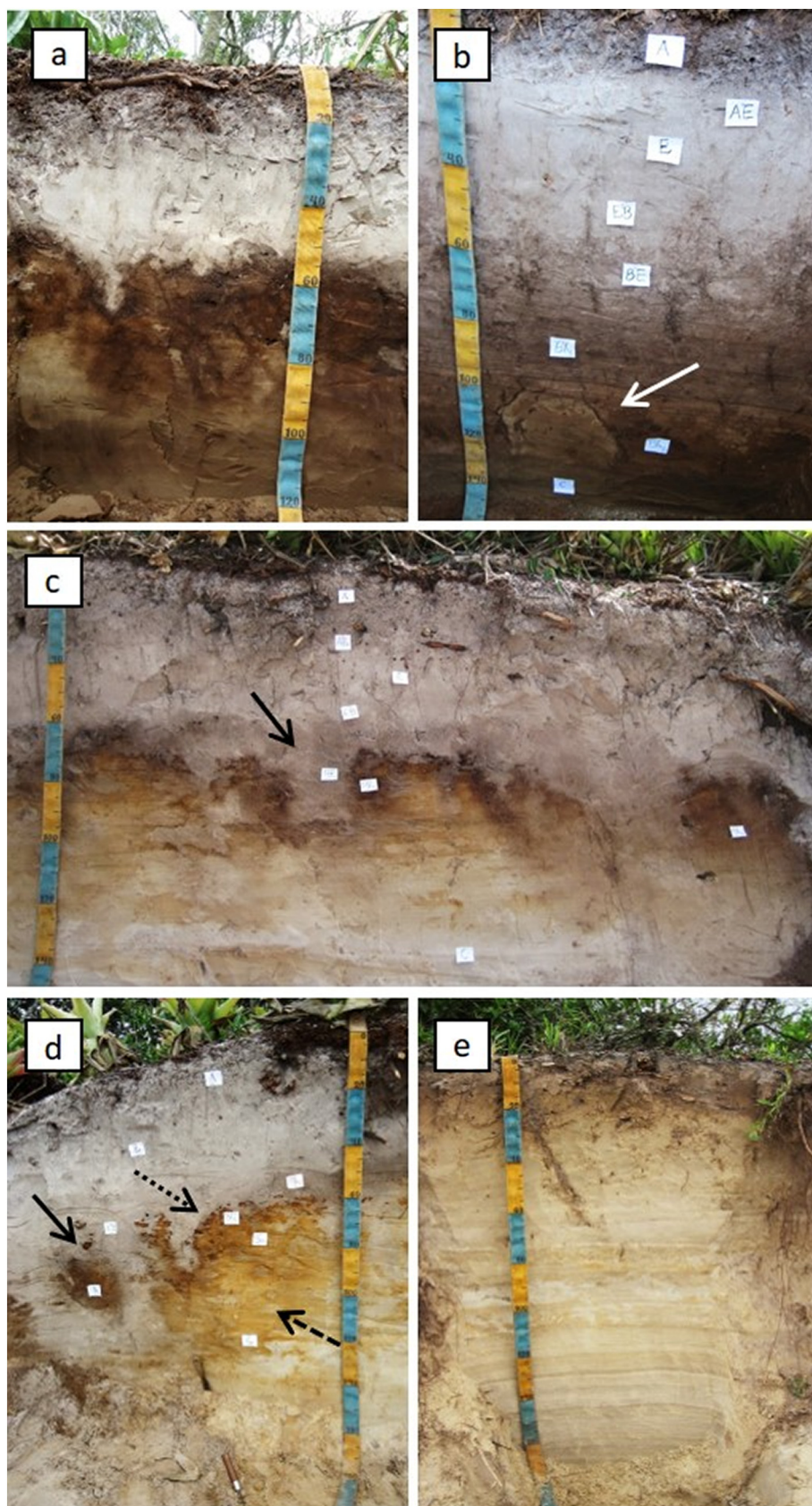


Fig. 8. (a) Profile P37. E horizon tongue and thin Bh. (b) Profile P38. Wide and irregular EB transition with abundant (decayed) roots, indicating drainage improvement. Irregular humus bands in Bh (arrow). (c) Profile P34. Discontinuous thin B horizon with Bs and tonguing E horizon. Remnants of decayed Bh horizon (arrow). (d) Profile P35. Note Bh island at left (arrow), presence of diffuse Fe compounds on right (dashed arrow) and Fe mottles at top of B horizon (dotted arrow). On the right, between 50 and 60 cm is a decayed Bh. (e) Profile P39. Arenosol without evidence of podzolisation. Rooted and stratified.

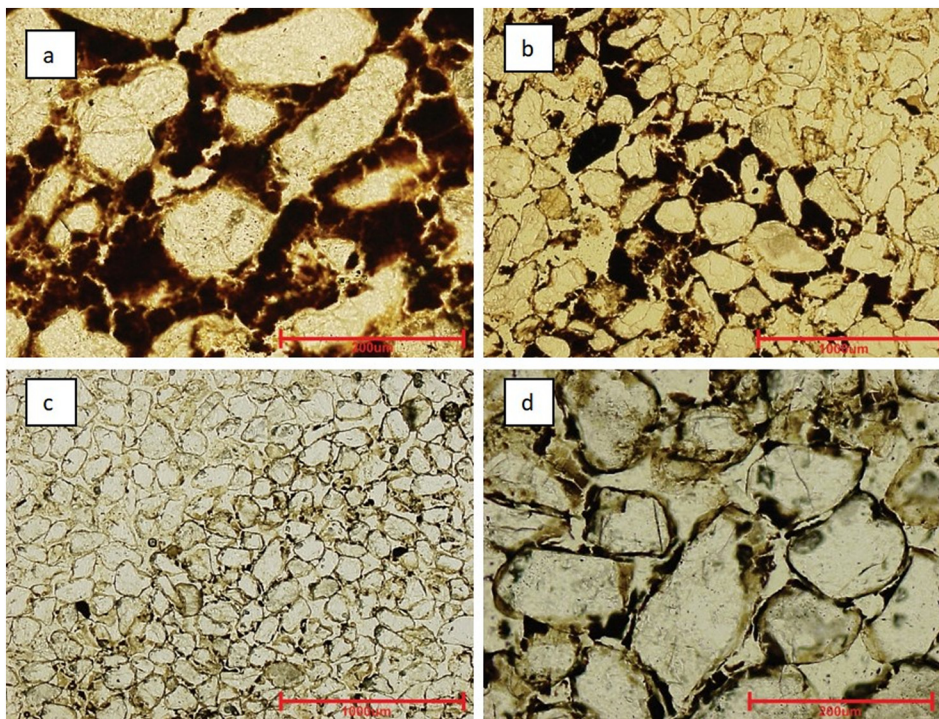


Fig. 9. OM coatings in thin sections from poorly drained podzols. (a) The Bhm₄ horizon from profile P4. Monomorphic OM completely fills the pores. Note the sharp boundaries and cracking of coatings. (b) Profile 4 (Bhm₄), a band cemented by monomorphic OM. (c) the EB₃ horizon of profile P2, showing remnants of degraded monomorphic coatings. (d) detail of (c).

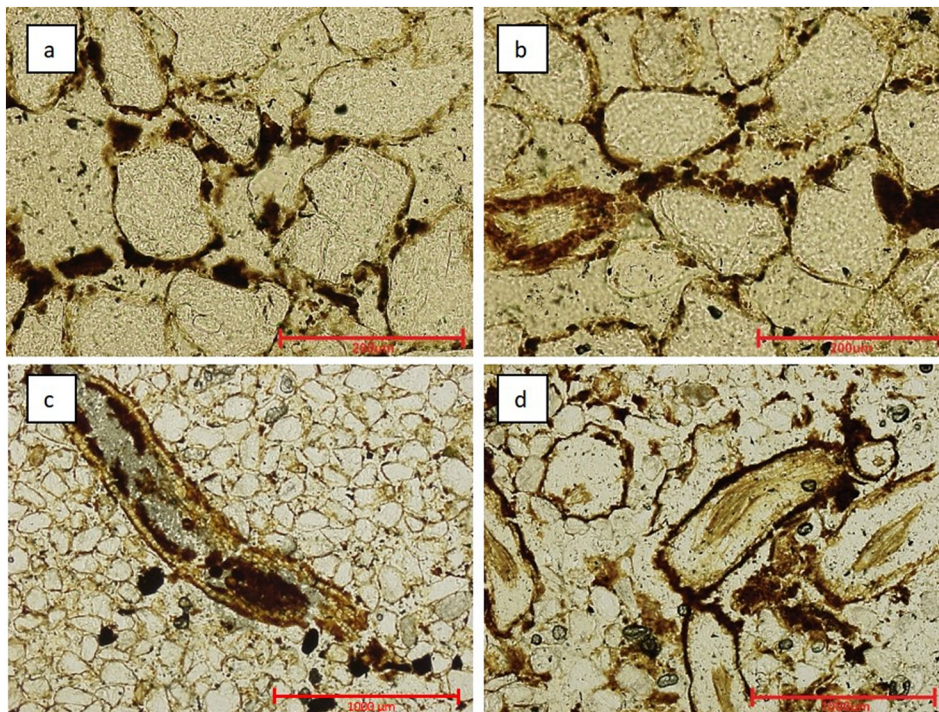


Fig. 10. Photomicrographs of poorly drained podzol P31: (a) decomposing monomorphic OM in the Bh₂ horizon, (b) the same horizon with decomposing root (centre left) and pellet-like OM, (c) decomposing roots, and (d) overview of decomposing roots in the Bh₂ of profile P38.

groundwater flow caused by converging ridges. (Fig. 1b; situation 3a in Fig. 2), in combination with gradual uplift of the SW part of the island (Martinez et al., 2018). This uplift also explains the degradation features encountered at the EB horizon transition in these profiles: uplift caused gradually improving drainage of the upper part of the profile. Thus, external lateral groundwater flow dominated podzol B horizon formation during the first two stages of landscape formation, while during subsequent stages, vertical fluctuation rather than lateral flow was dominant in the poorly drained profiles (see also Fig. 2). In Units III and IV, the poorly drained podzols had a thin B horizon. Given the fact that present groundwater fluctuation in these Units is compatible with

the morphology of the podzol profiles found in these units (Martinez et al., 2018), uplift and tilting will not have occurred after the formation of Unit II.

Although uplift only occurred until Unit II, improved drainage also occurred in originally poorly drained podzols from Units III and IV due to lowering of the erosion base close to the shore. The morphology of originally poorly drained podzols in Units I–IV showed degradation features related to improved drainage. This also affected profiles up to 100 m inland from the S cliff. Improved drainage implies that the profile is influenced by vertical percolation by rain water rather than by fluctuations of the groundwater table. Effects of improved drainage on

the morphology of originally hydromorphic profiles were described from the Netherlands and from SE Brazil. Such improved drainage leads to a gradual deepening of the E horizon accompanied by a fading EB boundary (Buurman et al., 2005, 2013). In the study area, this gradual deepening of the E was accompanied by formation of tonguing E horizons due to increased vertical water transport along abandoned root channels. In several cases, new humus bands were formed in E and EB horizons. These bands should be formed by precipitation of DOM from above.

The degradation process was accompanied by a number of localized phenomena of OM depletion (depletion mottles). Of these mottles, types a, c and d did not affect the sedimentary structures, which indicates that they are due to microbial degradation and not to burrowing. Type b and e mottles were accompanied by obliteration of sedimentary stratification and can therefore be interpreted as burrows and coalescent burrows, inside of which OM decay may have occurred. The sharp boundaries of types b and e support this interpretation. Type b depletions were virtually restricted to the EB and BE transitional horizons. This may imply that such burrows are formed exclusively in materials that have already lost Al due to increased vertical percolation. Type a, c and d depletions occurred within the apparently unburrowed hydromorphic Bh horizons and are thus not restricted by Al bound in OM complexes.

4. Conclusions

For the podzols of Ilha Comprida, drainage conditions at the time of podzol formation can be recognised morphologically. Poorly drained podzols with and without DOM accumulation through lateral groundwater flow, well-drained podzols and transitions between these were distinguished.

The poorly drained profiles on the oldest Geomorphic Units (Units I and II) showed various degradation features caused by improved drainage conditions. This supports uplift after the formation of Units I and II as was proposed by Martinez et al. (2018). On younger surfaces (Units III–V), the morphology of the inland podzols was still in equilibrium with present hydrology while those in the cliff also showed signs of degradation due to improved drainage.

Degradation features caused by improved drainage of originally poorly drained B horizons included (1) deepening of the E horizon along fossil root channels, causing tonguing and sometimes convoluted EB transitions due to preferential water flow, (2) a diffuse EB boundary where tonguing is absent, (3) different types of OM-depleted mottles, both within the EB horizon and throughout the B, (4) thin bands of (amorphous) OM, including i) horizontal bands and ii) bands that followed former root channels; the horizontal bands continued through tongue structures, independently of E or B material, which indicates precipitation of DOM from vertically percolating water and must have formed after both formation and degradation of the B horizon. The bands that follow root channels or E horizon tongues probably formed while the (decaying) root was still present.

Because morphological features in podzol E and B horizons reflect formation and degradation processes and may provide their relative chronology, they can be a valuable framework in interpreting chemical studies.

The rather large variation in B horizon morphology within a short distance demonstrates the relevance of studying transects instead of single profiles. Connecting the morphological interpretation of several podzol profiles provides insight into groundwater dynamics.

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